

Shane W. Morton

EDUCATION

University of Colorado Boulder | Boulder, CO
BS Mechanical Engineering

August 2025 – May 2028

Campus involvement

Formula SAE power train, Executive Board Alpha Tau Omega Fraternity, Physics Club Vice President

WORK & OTHER EXPERIENCE

CU Boulder ITL Laboratory, Boulder, CO

January 2026 – Present

Engineering Support Student

- Provided a technical support and customer service role in Boulder's Engineering Laboratory.
- Managed the maintenance and cleaning of 20 Prusa MK4S 3D printers and 5 Epilog laser cutters to optimize workflow and reduce print failure rates.
- Instructed students on 3D printer operation, material selection, and file preparation using SolidWorks and 3DPrinterOS.
- Mentored ~30 undergrads per shift with prototyping, manufacturing, and presentation for their first-year projects.

Parker Meggitt USA Inc., Simi Valley, CA

May 2025 – July 2025

Mechanical Engineering Intern

- Assisted engineers in the fire extinguishing unit with R&D tests for fire extinguisher bottles and cartridges to develop a new product line of outlets.
- Performed error analysis on ~100 defective hermetically sealed bottles to identify and correct errors in the manufacturing process.
- Led initiative to update and streamline outdated company documentation to improve readability and continuity.
- Studied manufacturing process to understand product evolution from CAD and sketching to packaging and shipping.

Morton Manufacturing, Lancaster, CA

June 2024 – August 2024

Calibration Intern

- Assisted with company-wide calibration, GAGE R&R, and uncertainty and bias testing to comply with the National Aerospace Standard and Aerospace Material Specification
- Optimized workflow for calibrating micrometers in the calibration department to efficiently check in and out measurement tools to manufacturing engineers with reduced delay.
- Calibrated 20 tools on average per shift
- Worked with micrometers, drop indicators, calipers, gauge blocks, and other aerospace fastener production tools to complete calibration responsibilities.

ACADEMIC PROJECTS

CU Boulder – Design, Production, and Presentation of Crowd Counting Device

Fall 2025

- Identified and analyzed problems with crowded spaces to devise our product.
- Collaborated in a multidisciplinary team of engineering students to develop a finished design.
- Took the lead on electrical component design, the coding process, and presentation.
- Iteratively designed and tested numerous prototypes, modifying for accuracy and efficiency.
- Presented final design at CU Boulder Project Expo to industry representatives, faculty, and peers.

SKILLS

- **CAD/Design** – SolidWorks (CSWA Certified), Autodesk Fusion 360, BrickLink Studio
- **Hardware and Fabrication** – 3D Printing, Soldering, Woodworking, Welding, Laser cutting, Lathes & Mills, Sewing
- **Programming** – Python, MATLAB, Arduino
- **Office Communication** – Microsoft Office, Smarteam, LibreOffice, Jira, Adobe Acrobat